W1(R) B_F (≥ 0). This quantity must be greater than or equal to 0.

W2(R) C_F (≥ 0). This quantity must be greater than or equal to 0.

W3(R) B_R (≥ 0). This quantity must be greater than or equal to 0.

W4(R) C_R (≥ 0). This quantity must be greater than or equal to 0.

W5(I) Junction number.

8.8.20 Pressurizer ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required of any of the interphase friction flags b in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

8.9 CANDU Channel Component, CANCHAN

A CANDU channel component is indicated by CANCHAN for Word 2 on CCC0000.

More than one junction may be connected to the inlet or outlet of a CANDU channel. If an end has no junctions, that end is considered a closed end. For major edits, minor edits, and plot variables, the volumes in the CANDU channel component are numbered as CCCNN0000, where NN is the volume number (01 ≤ NN ≤ 99). The junctions in the CANDU channel component are numbered as CCCJJ0000, where JJ is the junction number (01 ≤ JJ ≤ 98).

The volumes in a CANDU channel are usually considered one-dimensional components and flow in the volumes is along the x-coordinate. Crossflow junctions can connect to any of the CANDU channel volumes in the y- and z-coordinate directions (faces 3, 4, 5, or 6)using a form of the momentum equation that does or does not include momentum flux terms. It is also possible to connect external junctions to the x-coordinate direction faces (1 or 2) of any of the CANDU channel volumes using a form of the momentum equation that does or does not include the momentum flux terms. It is also possible to include or not include the momentum flux terms in internal CANDU channel junctions.

8.9.1 CANDU Channel Information, CCC0001

This card is required.

W1(I) Number of volumes, nv. The number nv must be greater than 0 and less than 100. The number of associated junctions internal to these components is nv-1. The outer junctions are described by other components.

W2(I) Surgeline connection number. This word must the same format as printed in the output.

W3(R) User specified interfacial heat transfer coefficient from liquid to saturation state (W/m2-K, Btu/hr-ft2-F). This word is optional.

W4(R) User specified interfacial heat transfer coefficient from vapor to saturation state (W/m2-K, Btu/hr-ft2-F). This word is optional.

8.9.2 CANDU Channel X-Coordinate Areas, CCC0101-0199

These cards are required, and the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv sets. (The corresponding y- and z-coordinate cards are CCC1601-1699 and CCC1701-1799). The words for one set are

W1(R) Area (m^2, ft^2).

W2(I) Volume number.

8.9.3 CANDU Channel Junction Areas, CCC0201-0299

These cards are optional, and, if entered, the card numbers need not be consecutive. The format is two words per set in sequential expansion format for nv-1 sets.

W1(R) Internal junction flow area (m^2, ft^2). If cards are missing or a word is 0, the junction flow area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or less than the minimum of the adjacent volume areas. There is no restriction for smooth area changes.

W2(I) Junction number.

8.9.4 CANDU Channel X-Coordinate Lengths, CCC0301-0399

These cards are required. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC1801-1899 and CCC1901-1999).

W1(R) Length (m, ft).

W2(I) Volume number.

8.9.5 CANDU Channel Volumes, CCC0401-0499

These cards are optional, and if not entered, the volumes are set to 0. The format is two words per set in sequential expansion format for nv sets. Card numbers need not be consecutive.

W1(R) Volume (m^3, ft^3). The code requires that each volume equal the x-direction flow area times the x-direction length. If activated, the code also requires each volume equal the y-direction flow area times the y-direction length, and each volume equal the z-direction flow area times the z-direction length. For any volume, at least two of the three quantities, x-direction area, the x-direction length, or volume, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the quantities are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions. If both the y-direction area and y-direction length are not entered or are 0, the y-direction length is computed from $2.0 \bullet \left(\frac{\text{x} - \text{direction area}}{\pi}\right)^{0.5}$ and the y-direction area is computed from $\frac{\text{volume}}{\text{y-direction length}}$. The same is true for the z-direction.

W2(I) Volume number.

8.9.6 CANDU Channel Volume Azimuthal Angles, CCC0501-0599

These cards are optional, and if not entered, the angles are set to 0. The angles on these cards are used for the 3-D drawing programs, so it is recommended that the user enter these values if they are known. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Azimuthal angle (degrees). The absolute value of the angle must be $\leq$ 360 degrees.

W2(I) Volume number.

8.9.7 CANDU Channel Volume Vertical Angles, CCC0601-0699

These cards are required. The format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(R) Vertical angle (degrees). The absolute value of the angle must be less than or equal to 90 degrees. This angle is used in the interphase drag calculation.

W2(I) Volume number.

8.9.8 CANDU Channel X-Coordinate (Elevation) Changes, CCC0701-0799

These cards are optional. If these cards are missing, the coordinate changes or elevation changes are computed from the x-coordinate length and a rotation matrix computed from the angle information. If these cards are entered, the entered data becomes the x-coordinate change or elevation change data. Two formats entering 1 or 3 coordinate changes per volume are provided. The card format is two or four words per set in sequential expansion format up to nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2101-2199 and CCC2201-2299).

1 coordinate change per volume format:

W1(R) Elevation change. This is the coordinate change along the fixed x-axis due to the traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft). A positive value is an increase in elevation. The magnitude must be equal to or less than the length. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used; when the ratio is greater than 45 degrees, the vertical flow regime map is used.

W2(I) Volume number.

Three coordinate changes per volume format:

W1(R) Coordinate change along the fixed x-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{xx} (m, ft).

W2(R) Coordinate change along the fixed y-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{yx} (m, ft).

W3(R) Coordinate change along the fixed z-axis due to traverse from inlet to outlet along the local x-coordinate, Δ_{zx} (m, ft).

W4(I) Volume number.

8.9.9 CANDU Channel Volume X-Coordinate Friction Data, CCC0801-0899

These cards are required. The card format is three words per set for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2301-2399 and CCC2401-2499).

W1(R) Wall roughness (m, ft).

W2(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$.

If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made to ensure that the roughness is less than half the hydraulic diameter. See Word 1 on cards CCC0101-0109 for the area.

W3(I) Volume number.

8.9.10 CANDU Channel Junction Loss Coefficients, CCC0901-0999

These cards are optional and if missing, the energy loss coefficients are set to 0. The card format is three words per set in sequential expansion format for nv-1 sets, and card numbers need not be consecutive.

W1(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. Note: a variable loss coefficient may be specified (see Section 8.6.19). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected for this junction on cards CCC1101-1199. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area (from cards CCC0201-0299). If one of the abrupt area change options is selected (a = 1 or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction (from cards CCC0101-0199). If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

W2(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. Note: a variable loss coefficient may be specified (see Section 8.6.19). See information for the previous word regarding interpretation and use of this loss coefficient.

W3(I) Junction number.

8.9.11 CANDU Channel Volume X-Coordinate Control Flags, CCC1001-1099

These cards are required. The card format is two words per set in sequential expansion format for nv sets, and card numbers need not be consecutive. (The corresponding y- and z-coordinate cards are CCC2701-2799 and CCC2801-2899).

W1(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. These words enter the scalar oriented flags and the x-coordinate flags for each volume in the component.

t-flag. The digit t specifies whether the thermal front tracking model is to be used; t = 0 specifies that the front tracking model is not to be used for the volume, and t = 1 specifies that the front tracking model is to be used for the volume. The thermal front tracking model can only be applied to vertically-oriented components.

l-flag The digit l specifies whether the mixture level tracking model is to be used; l = 0 specifies that the level model not be used for the volume, and l = 1 specifies that the level model be used for the volume. The mixture level tracking model can only be applied to vertically-oriented components.

p-flag The digit p specifies whether the water packing scheme is to be used. p = 0 specifies that the water packing scheme is to be used for the volume, and p = 1 specifies that the water packing scheme is not to be used for the volume.

v-flag The digit v specifies whether the vertical stratification model is to be used. v = 0 specifies that the vertical stratification model is to be used for the volume, and v = 1 specifies that the vertical stratification model is not to be used for the volume.

b-flag The digit b specifies the interphase friction that is used. b = 0 means that the CANDU channel interphase friction model will be applied, b = 1 means that the rod bundle interphase friction model will be applied, b = 2 means that the ORNL ANS interphase friction model will be applied (see card ccc3101 in Section 8.6.20).

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and f = 1 specifies that wall friction effects are not to be computed along the x-coordinate.

e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

W2(I) Volume number.

8.9.12 CANDU Channel Junction Control Flags, CCC1101-1199

These cards are required. The card format is two words per set in sequential expansion format for nv–1 sets, and card numbers need not be consecutive.

W1(I) Junction control flags. This word has the format jefvcahs. It is not necessary to input leading zeros. For the CANDU channel component, the junction control flag is limited to the format 0ef0cahs.

e-flag The digit e specifies the modified PV term in the energy equations. e = 0 means that the modified PV term will not be applied, and e = 1 means that it will be applied.

f-flag The digit f specifies CCFL options. f = 0 means that the CCFL model will not be applied, and f = 1 means that the CCFL model will be applied.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and contains Option 50, the original RELAP5 critical flow model is active. Unlike junctions for certain other hydrodynamic components, only the default values of discharge coefficients and factors used in the critical flow models are allowed for junctions in CANDU channel components. Problem renodalization should be used to circumvent this limitation should it prove to be significant. For the Henry-Fauske critical flow model, the default value for the discharge coefficient is 1.0 and the default value for the thermal nonequilibrium constant is 0.14. For the original RELAP5 critical flow model, the default values for the subcooled, two-phase and superheated discharge coefficients each are 1.0.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as ¨a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option, and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show a 1.

s-flag The digit s specifies momentum flux options. s = 0 uses momentum flux in both the to volume and the from volume. s = 1 uses momentum flux in the from volume, but not in the to volume. s = 2 uses momentum flux in the to volume but not in the from volume. s = 3 does not use momentum flux in either the to or the from volume. For this component, the option s = 0 is the usual recommendation (momentum flux in both volumes). The other options s = 1, 2, and 3 are included to allow consistency for this flag for other components (single-junction, branch junction, etc.).

W2(I) Junction number.

8.9.13 CANDU Channel Volume Initial Conditions, CCC1201-1299

These cards are required. The card format is seven words per set in sequential expansion format for nv sets, and card numbers need not be consecutive.

W1(I) Control word. This word has the format εbt. It is not necessary to input leading zeros.

e-flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., ε = 1 specifies H_2O, ε = 2 specifies D_2O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default ε = 0, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.

b-flag The digit b specifies whether boron is present or not. b = 0 specifies that the volume liquid does not contain boron; b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered on the CCC2001-2099 cards.

t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. Entering t equal to 0-3 specifies one component (steam/water). Entering t equal to 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

$[P,U_f,U_g,\alpha_g]$ If t = 0, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions, depending on the internal energies used to define the thermodynamic state. W6 should be 0.0.

One Component (Steam/Water), Equilibrium

[T,x_s] If t = 1, the next two words are interpreted as temperature (K, ^oF) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,x_s] If t = 2, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. W4, W5, and W6 should be 0.0.

[P,T] If t = 3, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, ^oF) in equilibrium condition. W4, W5, and W6 should be 0.0.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E\text{-}9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\ E\text{-}9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If t = 4, the next three words are interpreted as pressure (Pa, lb_f/in^2), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and ≤ 1.0, saturated noncondensables will result. W5 and W6 should be 0.0. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

[T,x_s,x_n] If t = 5, the next three words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. W5 and W6 should be 0.0. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g,α_g,x_n] If t = 6, the next five words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing (t = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy.

W2-W6(R) Quantities as described under Word 1. Five quantities must be entered, and zeros should be entered for unused quantities. If any control word (Word 1) indicates that boron is present, cards CCC2001-2099 must be entered to define the initial boron concentrations. Boron concentrations are not entered in Words 2-6 for CANDU channel components.

W7(I) Volume number.

8.9.14 CANDU Channel Junction Conditions Control Words, CCC1300

This card is optional, and, if missing, velocities are assumed on cards CCC1301-1399.

W1(I) Control word. If 0, the first and second words of each set on cards CCC1301-1399 are velocities. If 1, the first and second words of each set on cards CCC1301-1399 are mass flows.

8.9.15 CANDU Channel Junction Initial Conditions, CCC1301-1399

These cards are required.

W1(R) Initial liquid velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). This capability has not yet been implemented, so enter 0.

W4(I) Junction number.

8.9.16 CANDU Channel Junction Diameter and CCFL Data, CCC1401-1499

These cards are optional. The defaults indicated for each word are used if the card is not entered. If this card is being used to specify only the junction hydraulic diameter for the interphase drag calculation, (i.e., f = 0 in Word 1 of cards CCC1101-1199) then the diameter should be entered in Word 1 and any allowable values should be entered in Words 2-4 (will not be used). If this card is being used for the CCFL model (i.e., f = 1 in Word 1 of cards CCC1101-1199), then enter all four words for the appropriate CCFL model if values different from the default value are desired.

W1(R) Junction hydraulic diameter, D_j (m, ft). This quantity is the junction hydraulic diameter used in the CCFL correlation equation and interphase drag and must be ≥ 0. The number should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$. If a 0 is entered or if the default is used, the junction diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. See Word 1 of cards CCC0201-0299 for the area.

W2(R) Flooding correlation form, β. If 0, the Wallis CCFL form is used. If 1, the Kutateladze CCFL form is used. If between 0.0 and 1.0, Bankoff weighting between the Wallis and Kutateladze CCFL forms is used. This number must be ≥ 0 and ≤ 1. The default value is 0 (Wallis form).

W3(R) Gas intercept, c. This quantity is the gas intercept used in the CCFL correlation (when $H_f^{1/2} = 0$) and must be > 0. The default value is 1.

W4(R) Slope, m. This quantity is the slope used in the CCFL correlation and must be > 0. The default value is 1.

W5(I) Junction number.

8.9.17 CANDU Channel Initial Boron Concentrations, CCC2001-2099

These cards are required only if boron is specified in one of the control words (Word 1) in cards CCC1201-1299. The card format is two words per set in sequential expansion format for nv sets. Boron concentrations must be entered for each volume, and 0 should be entered for those volumes whose associated control word did not specify boron.

W1(R) Boron concentration (mass of boron per mass of liquid).

W2(I) Volume number.

8.9.18 CANDU Channel Volume Additional Wall Friction Data, CCC2501-2599

These cards are optional. If these cards are not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. The card format is seven words per set in sequential expansion format for nv sets and card numbers need not be consecutive.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

W7(I) Volume number.

8.9.19 CANDU Channel Junction Form Loss Data, CCC3001-3099

These cards are optional. The user-specified form loss is given in Words 1 and 2 of cards CCC0901-0999 if this card is not entered. If this card is entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 1 and 2 of cards CCC0901-0999. Re is the Reynolds number based on mixture fluid properties. If this card is being used for the form loss calculation, then enter all five words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R; the same information applies for B_F and B_R.

W1(R) B_F (≥ 0). This quantity must be greater than or equal to 0.

W2(R) C_F (≥ 0). This quantity must be greater than or equal to 0.

W3(R) B_R (≥ 0). This quantity must be greater than or equal to 0.

W4(R) C_R (≥ 0). This quantity must be greater than or equal to 0.

W5(I) Junction number.

8.9.20 CANDU Channel ORNL ANS Interphase Model Values, CCC3101-3199

These cards are required of any of the interphase friction flags b in the volume control flags entered on cards CCC1001-1099 are set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

8.10 Branch Component, BRANCH

A branch component is indicated by BRANCH for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N is the face number. More than one junction may be connected to the inlet or outlet. If an end has no junctions, that end is considered a closed end. If more than one junction is connected on one end of a branch, each junction should be modeled as an abrupt area change. For major edits, minor edits, and plot variables, the volume in the branch component is numbered as CCC010000. The junctions associated with the branch component are numbered as CCCJJ0000, where JJ is the junction number ($01 \leq JJ \leq 09$).

8.10.1 Branch Component Information, CCC0001

This card is required.

W1(I) Number of junctions, nj. The variable nj is the number of junctions described in the input data for this component ($0 \leq nj \leq 9$). Not all junctions connecting to the branch need be described with this component input, and nj is not necessarily the total number of junctions connecting to the branch. Junctions described in single-junctions, time-dependent junctions, pumps, circulators, separators, jetmixers, ECC mixers, and other branches can be connected to this branch.

W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.